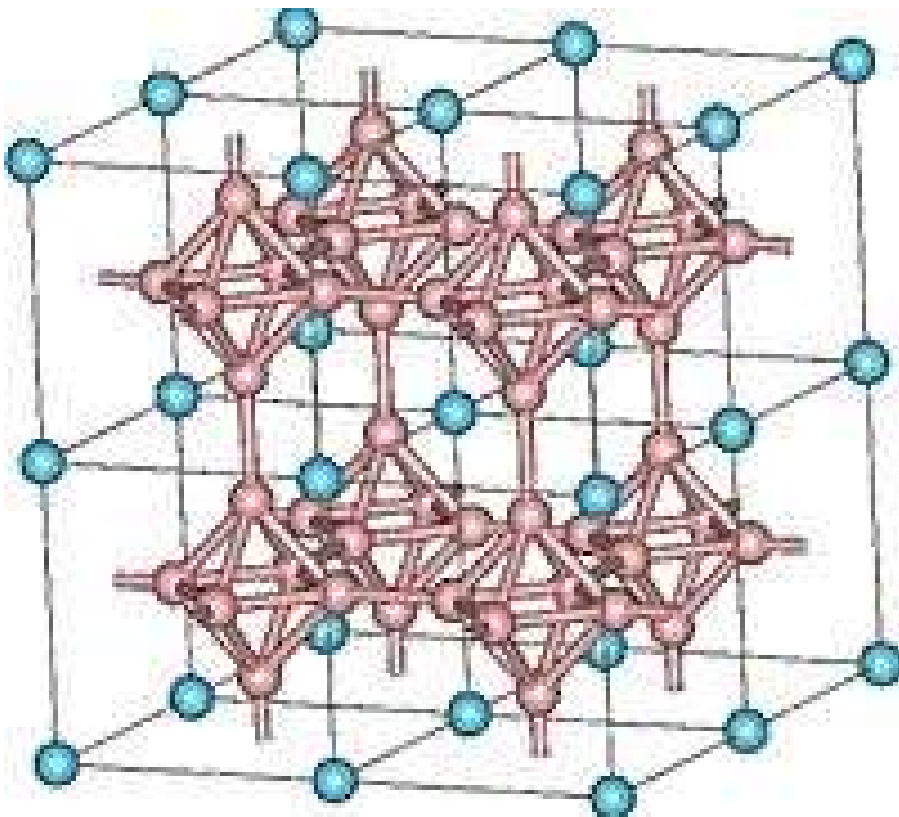


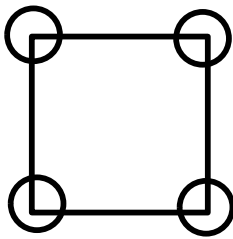
Hexaborides (CaB₆, LaB₆ etc.)

[excellent electron sources, generate large quantity of electrons when heated]

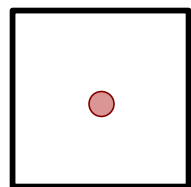


$Ca \rightarrow 000$

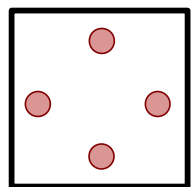
$B \rightarrow \begin{smallmatrix} 1 & 1 & 1 \\ 2 & 2 & 4 \end{smallmatrix} \quad \begin{smallmatrix} 1 & 1 & 3 \\ 2 & 2 & 4 \end{smallmatrix} \quad \begin{smallmatrix} 1 & 1 & 1 \\ 2 & 4 & 2 \end{smallmatrix} \quad \begin{smallmatrix} 1 & 1 & 1 \\ 4 & 2 & 2 \end{smallmatrix} \quad \begin{smallmatrix} 1 & 3 & 1 \\ 2 & 4 & 2 \end{smallmatrix} \quad \begin{smallmatrix} 3 & 1 & 1 \\ 4 & 2 & 2 \end{smallmatrix}$



$Z = 0, 1$



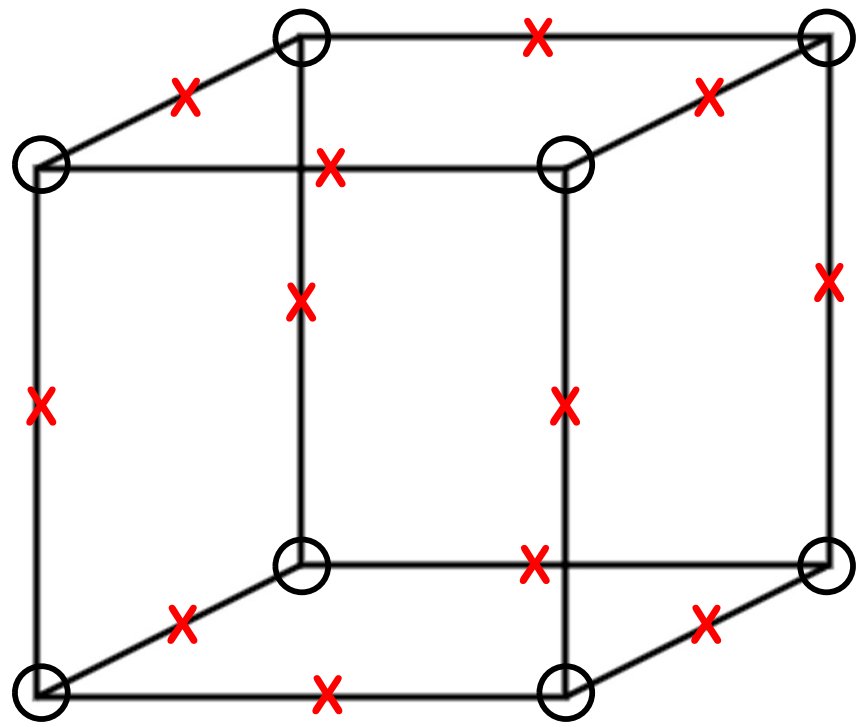
$Z = \frac{1}{4}, \frac{3}{4}$



$Z = \frac{1}{2}$

ReO₃

[BX₃ type]



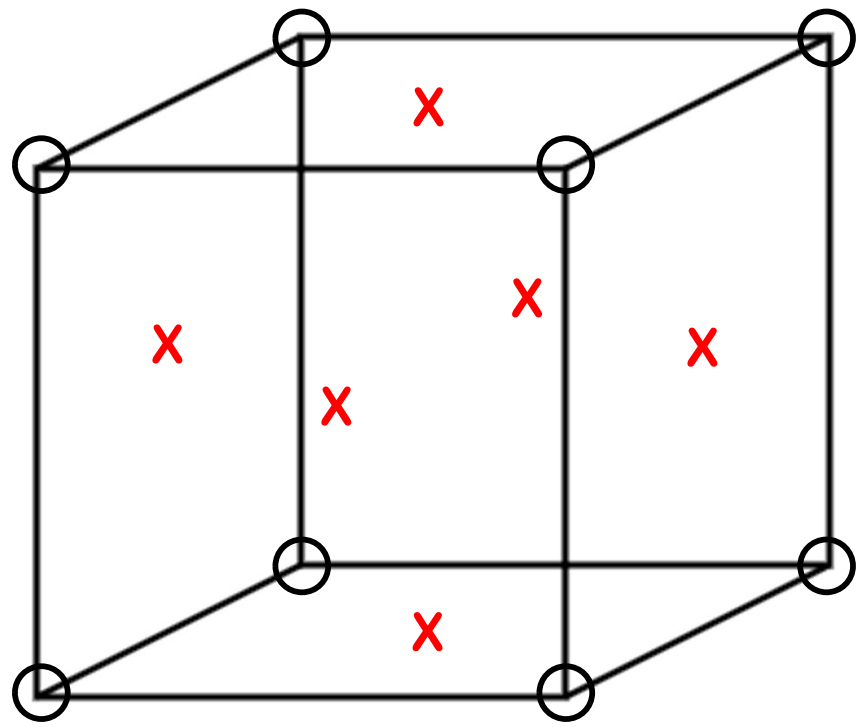
○ *Re* → 000

× *O* → $\frac{1}{2}00$ $0\frac{1}{2}0$ $00\frac{1}{2}$

Effective number of oxygen is 3

Cu₃Au

[AB₃ type]



Au → 000

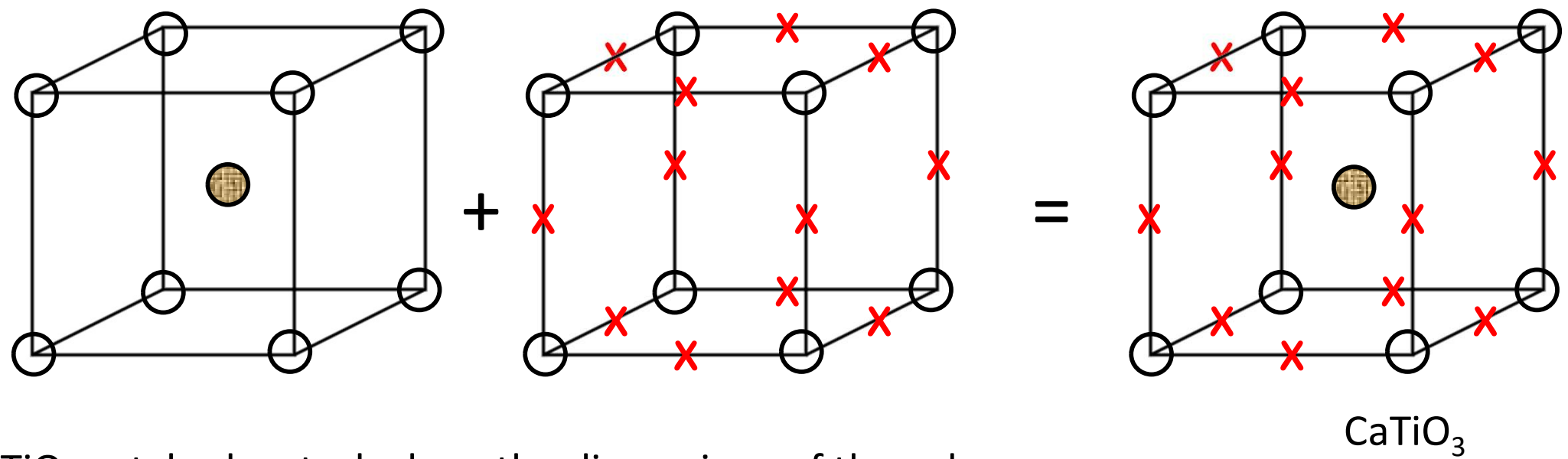


Cu → $\frac{1}{2}\frac{1}{2}0$ $0\frac{1}{2}\frac{1}{2}$ $\frac{1}{2}0\frac{1}{2}$

Putting another atom in each face center

CsCl + ReO₃

AB + BX₃



TiO₆ octahedra stack along the dimensions of the cube

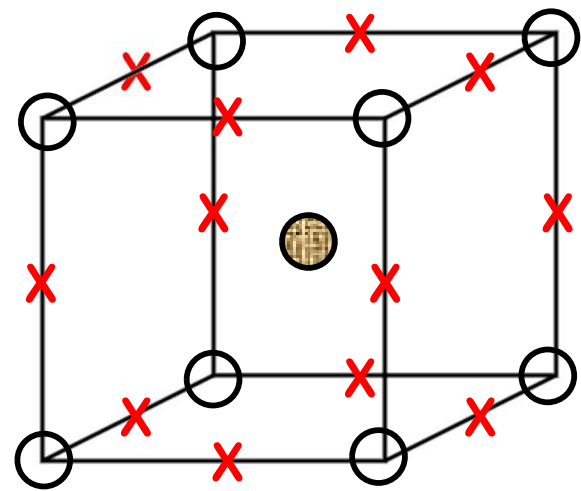
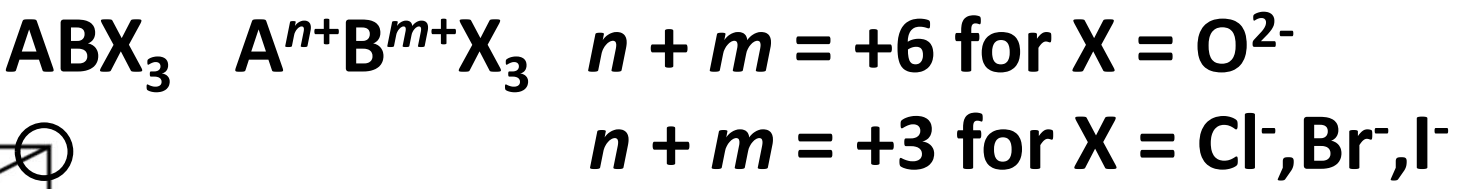
- CN of Ti → 6 (oxygen)
- CN of Ca → 12 (oxygen) [CaO₁₂] dodecahedra
- Oxygen occupy the octahedral voids

$Ti \rightarrow 000$

$Ca \rightarrow \frac{1}{2} \frac{1}{2} \frac{1}{2}$

$O \rightarrow \frac{1}{2} 00 \quad 0 \frac{1}{2} 0 \quad 00 \frac{1}{2}$

Perovskite Structure



- CaTiO₃
- SrTiO₃
- BaTiO₃
- LaCoO₃
- KNbO₃

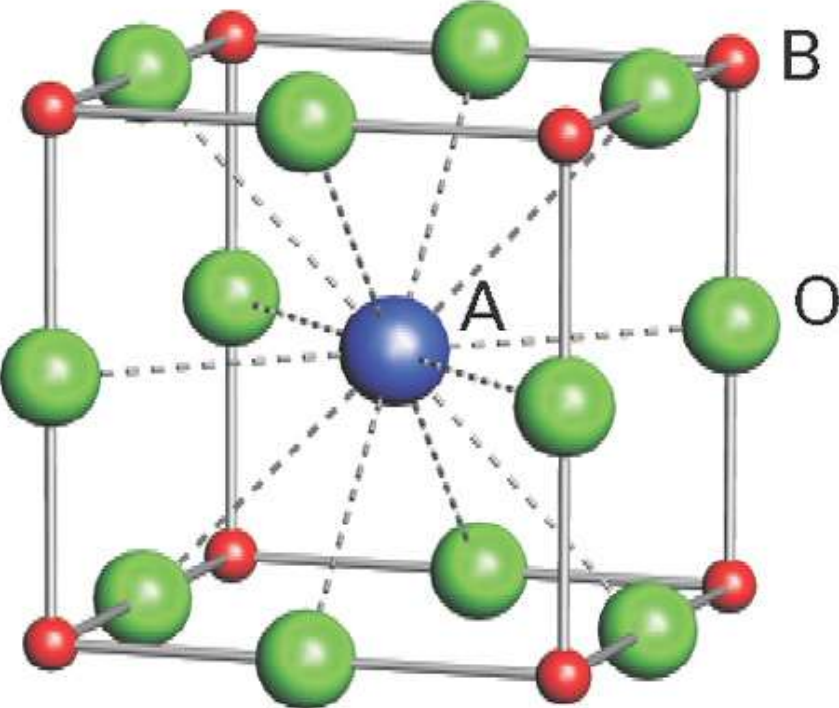
- CH₃NH₃PbI₃
- CsPbBr₃

Goldsmith Tolerance Factor (*t*)

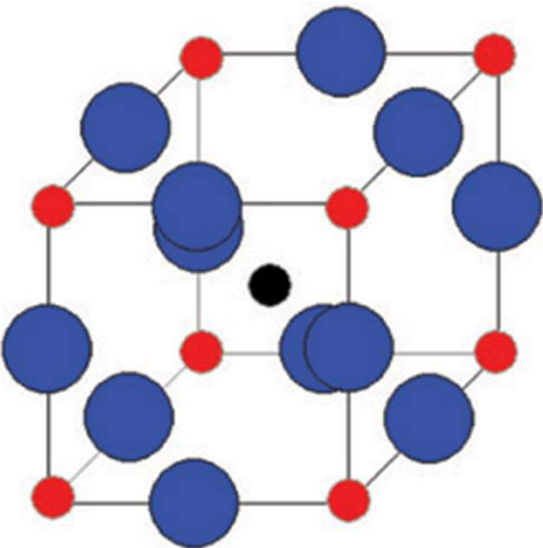
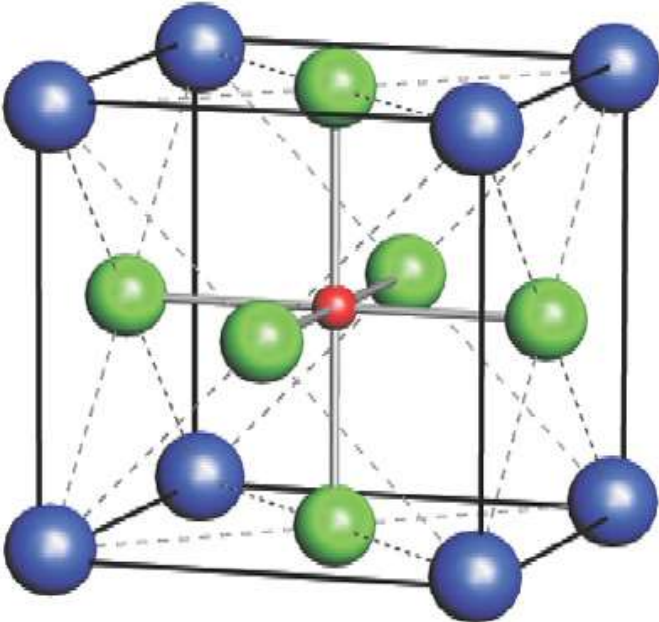
$$t = \frac{r_A + r_X}{\sqrt{2} (r_B + r_X)}$$

***t* = 1 (ideal perovskite)**
≈ ± 15 %

Perovskite Structure



This is the other way round

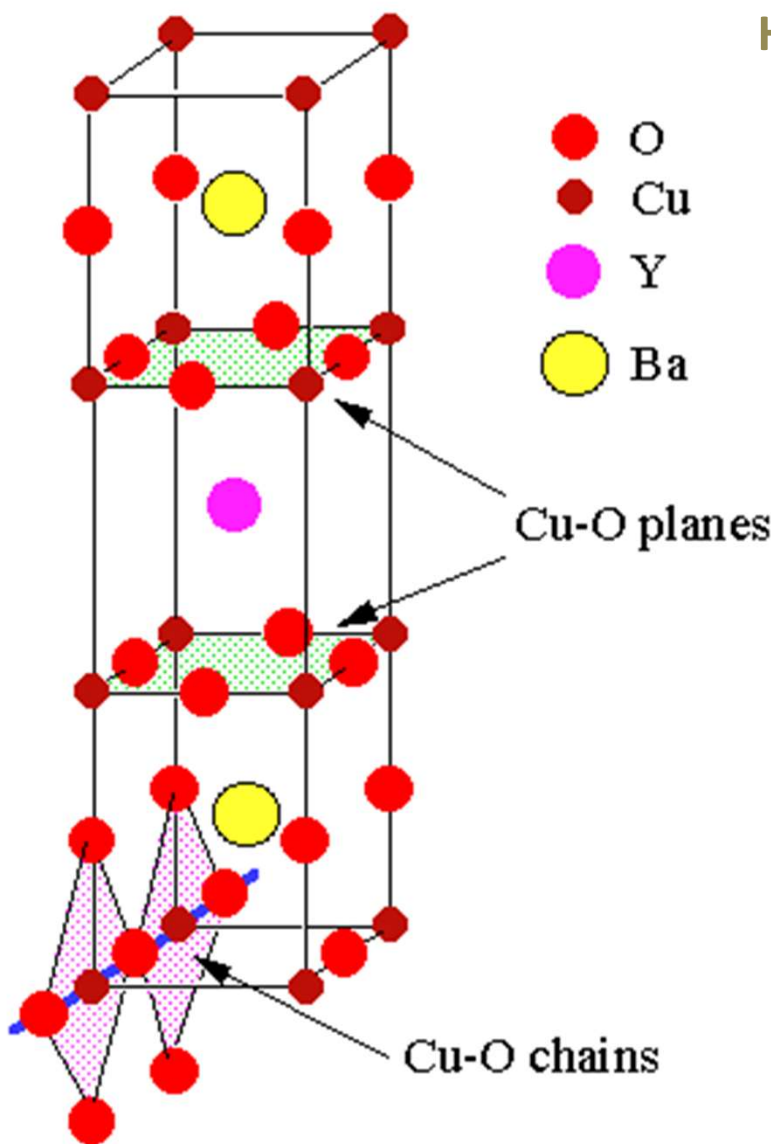


- Barium atom
- Titanium atom
- Oxygen atom

Related Structure

$\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$

High T_c superconductor



13 (+)charges
14 (-) charges

If δ is 0.5, structure will be neutral with 6.5 O
(Heat in vacuum)

If $\delta < 0.65$, orthorhombic
If $\delta = 1$, tetragonal

Oxygen non-stoichiometry

- $\text{Cu}^{2+}/\text{Cu}^{3+}$
- Ba^{2+}
- Y^{3+}
- × O^{2-}

